

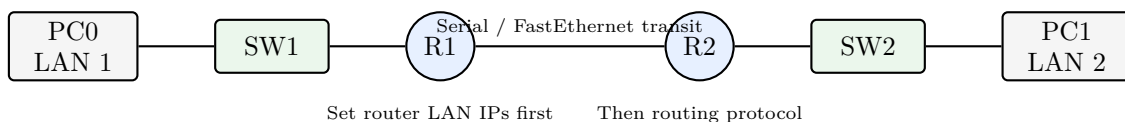
Networking Eval Cheatsheet

Lab 10 (RIP/OSPF) + Lab 11 (DHCP/NAT/NTP)

Built from your RIP, OSPF, NAT, DHCP, help notes, and lab manual material.

0. Zero-Panic Order: Do This First Every Time

1. Draw the topology and make an IP table before typing commands.
2. Unless the question already gives addresses, use /24 for LANs and /30 for router-to-router serial links.
3. Put the router interface IP first, then the PC IPs, then the default gateway on each PC.
4. On every router interface: assign IP, then **no shutdown**.
5. On a serial DCE side only: add **clock rate 64000**. If serial ports are missing in Packet Tracer, power off the router and add a WIC-2T module.
6. Verify base links with **show ip interface brief** before enabling RIP, OSPF, DHCP, or NAT.
7. After base connectivity works, add routing. After routing works, add DHCP or NAT.
8. Test in this order: own gateway → next-hop router → remote router LAN interface → remote PC.
9. Save at the end: **copy running-config startup-config**



1. Safe IP Assignment Without Bricking the Topology

1.1 Quick Addressing Rules

CIDR	Mask	Wildcard	Usable Hosts	Best Use
/24	255.255.255.0	0.0.0.255	254	Normal LAN with PCs
/30	255.255.255.252	0.0.0.3	2	Router-to-router point-to-point link
/8	255.0.0.0	0.255.255.255	16,777,214	Old class-A style lab questions

1.2 Default Gateway Rule

Every PC's default gateway is the IP address of the router interface on the same LAN.

Device	IP	Mask	Default Gateway
R1 LAN interface	192.168.10.1	255.255.255.0	—
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1

1.3 Private IP Ranges

- 10.0.0.0 - 10.255.255.255
- 172.16.0.0 - 172.31.255.255
- 192.168.0.0 - 192.168.255.255

1.4 Fast /30 Subnet Trick

Use this when you see a router-to-router link.

- Block size = $256 - \text{last octet of subnet mask}$
- For /30, mask = 255.255.255.252, so block size = $256 - 252 = 4$
- Blocks go as .0, .4, .8, .12, ...

Example: IP = 192.168.1.10/30

- Closest lower block = 192.168.1.8
- Network = 192.168.1.8
- Usable = 192.168.1.9, 192.168.1.10
- Broadcast = 192.168.1.11

1.5 Safe Addressing Template if Question Gives You Freedom

Segment	Example Network	Suggested Device IPs
Left LAN	192.168.10.0/24	Router = 192.168.10.1, PCs = .2, .3, ...
R1-R2 link	10.10.10.0/30	R1 = 10.10.10.1, R2 = 10.10.10.2
Right LAN	192.168.20.0/24	Router = 192.168.20.1, PCs = .2, .3, ...

2. Base Bring-Up Commands You Will Use in Almost Every Question

2.1 General Router Skeleton

```
enable
configure terminal
hostname R1
interface fa0/0
  ip address 192.168.10.1 255.255.255.0
  no shutdown
exit
interface s0/0/0
  ip address 10.10.10.1 255.255.255.252
  clock rate 64000
  bandwidth 64
  no shutdown
end
show ip interface brief
```

2.2 Packet Tracer Reminders

- Router to switch: usually copper straight-through.
- Router to PC directly: old labs may use cross-over, but Packet Tracer auto-cable usually works.
- Router to router serial: use serial DCE cable; set `clock rate 64000` only on DCE side.
- To check DCE side: `show controllers serial 0/0/0`
- Interface names vary: fa0/0, fa1/0, g0/0, s0/0/0, etc. Use whatever your device actually has.

2.3 Quick Verification Commands

Command	What it tells you
show ip interface brief	Fast check of all interface IPs and up/down status
show running-config	Full current config
show ip route	Routing table; look for C, R, O, S
ping x.x.x.x	Reachability check
copy running-config startup-config	Save your work

3. Static Routing: Good Backup If Dynamic Routing Is Not Asked

3.1 Command Pattern

```
ip route <destination-network> <mask> <next-hop-ip>
```

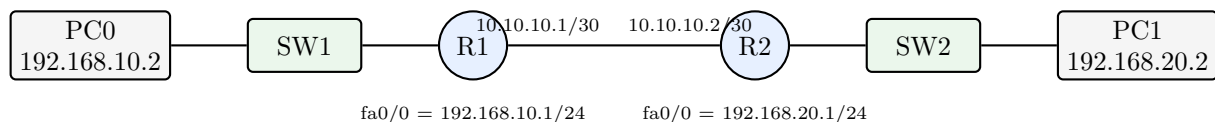
Example:

```
R1(config)# ip route 192.168.20.0 255.255.255.0 10.10.10.2
R2(config)# ip route 192.168.10.0 255.255.255.0 10.10.10.1
```

3.2 When to Use

- Small two-router questions
- DHCP relay questions that need routing but do not explicitly ask for RIP/OSPF
- If RIP/OSPF is broken and you need a quick fallback to prove connectivity

4. If the Question Says: “Add 2 Routers Using RIP”



4.1 PC Addressing

- PC0: IP 192.168.10.2, Mask 255.255.255.0, Gateway 192.168.10.1
- PC1: IP 192.168.20.2, Mask 255.255.255.0, Gateway 192.168.20.1

4.2 R1 Configuration

```
enable
configure terminal
hostname R1
interface fa0/0
  ip address 192.168.10.1 255.255.255.0
  no shutdown
exit
interface s0/0/0
  ip address 10.10.10.1 255.255.255.252
  clock rate 64000
  no shutdown
exit
router rip
  version 2
  no auto-summary
  network 192.168.10.0
  network 10.10.10.0
end
```

4.3 R2 Configuration

```
enable
configure terminal
hostname R2
interface fa0/0
  ip address 192.168.20.1 255.255.255.0
  no shutdown
exit
interface s0/0/0
  ip address 10.10.10.2 255.255.255.252
  no shutdown
exit
router rip
  version 2
  no auto-summary
  network 192.168.20.0
  network 10.10.10.0
end
```

4.4 RIP Notes That Save Marks

- RIP metric = hop count; maximum is 15 hops.
- Updates are sent every 30 seconds.
- If Packet Tracer is picky on a 10.10.10.0/30 serial link, try **network 10.0.0.0** instead of **network 10.10.10.0**.
- If the lab gives class-A style networks like 10.0.0.0, 20.0.0.0, 30.0.0.0, use those exact network numbers.

4.5 RIP Verification

```
show ip protocols
show ip route
ping 192.168.20.2
```

Look for R routes in `show ip route`.

5. If the Question Says: “Using OSPF”

5.1 Same 2-Router Topology, OSPF Area 0

Use the exact same addressing as the RIP example and only change the routing section.

5.2 OSPF Wildcard Rules

Mask	CIDR	Wildcard	Use
255.255.255.0	/24	0.0.0.255	LAN
255.255.255.252	/30	0.0.0.3	Router-to-router link
255.0.0.0	/8	0.255.255.255	Old class-A lab networks

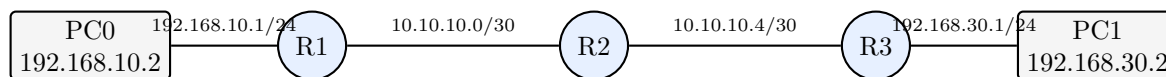
5.3 R1 OSPF Block

```
router ospf 1
  network 192.168.10.0 0.0.0.255 area 0
  network 10.10.10.0 0.0.0.3 area 0
```

5.4 R2 OSPF Block

```
router ospf 1
  network 192.168.20.0 0.0.0.255 area 0
  network 10.10.10.0 0.0.0.3 area 0
```

5.5 Full OSPF Example for 3 Routers (Close to the Lab Style)



All OSPF interfaces in area 0

```
R1
interface fa0/0
  ip address 192.168.10.1 255.255.255.0
  no shutdown
interface s0/0/0
  ip address 10.10.10.1 255.255.255.252
  clock rate 64000
  no shutdown
router ospf 1
  network 192.168.10.0 0.0.0.255 area 0
  network 10.10.10.0 0.0.0.3 area 0
```

```
R2
interface s0/0/0
  ip address 10.10.10.2 255.255.255.252
  no shutdown
interface s0/0/1
  ip address 10.10.10.5 255.255.255.252
  no shutdown
router ospf 1
  network 10.10.10.0 0.0.0.3 area 0
  network 10.10.10.4 0.0.0.3 area 0
```

```
R3
interface s0/0/1
  ip address 10.10.10.6 255.255.255.252
  no shutdown
interface fa0/0
  ip address 192.168.30.1 255.255.255.0
  no shutdown
router ospf 1
  network 10.10.10.4 0.0.0.3 area 0
  network 192.168.30.0 0.0.0.255 area 0
```

5.6 OSPF Verification

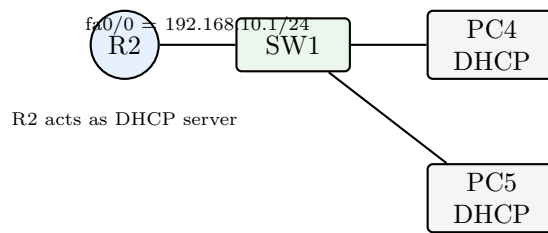
```
show ip ospf neighbor
show ip protocols
show ip route
ping 192.168.30.2
```

5.7 OSPF Mistakes to Check Fast

- Wrong wildcard mask
- Wrong area number
- Interfaces still down
- Hello/dead timers do not match

6. If the Question Says: “Router to Device Using DHCP”

6.1 Direct DHCP From Router to PCs on Same LAN



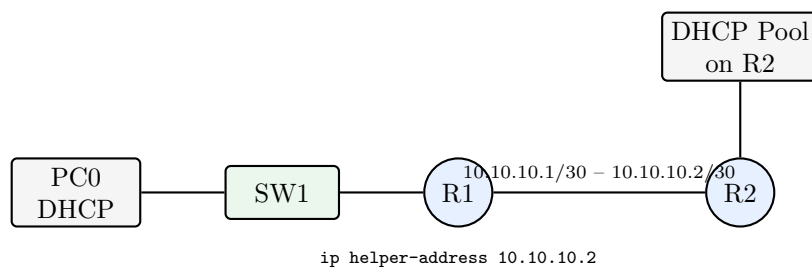
```
enable
configure terminal
hostname R2
interface fa0/0
  ip address 192.168.10.1 255.255.255.0
  no shutdown
exit
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp pool LAN1
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
  dns-server 8.8.8.8
end
```

Then set the PC IP configuration to DHCP.

6.2 DORA Process

- Discover
- Offer
- Request
- Acknowledge

6.3 DHCP Relay: Server and Clients on Different Routers



```
R2(config)# ip dhcp excluded-address 192.168.30.1 192.168.30.20
R2(config)# ip dhcp pool LAN30
R2(dhcp-config)# network 192.168.30.0 255.255.255.0
R2(dhcp-config)# default-router 192.168.30.1
R2(dhcp-config)# dns-server 8.8.8.8
```

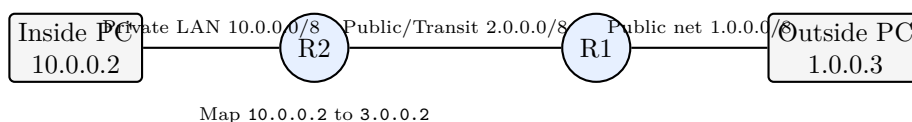
On the router interface facing the clients:

```
R1(config)# interface fa0/0
R1(config-if)# ip helper-address 10.10.10.2
```

Routing must already work between R1 and R2 using static routes, RIP, or OSPF.

7. If the Question Says: “NAT”

7.1 Static NAT (Matches Your Lab Material)



7.2 Router Config for Static NAT

```
R2(config)# ip route 1.0.0.0 255.0.0.0 2.0.0.1
R2(config)# ip nat inside source static 10.0.0.2 3.0.0.2
R2(config)# ip nat inside source static 10.0.0.3 3.0.0.3
R2(config)# interface fa0/0
R2(config-if)# ip nat inside
R2(config-if)# exit
R2(config)# interface s0/0/0
R2(config-if)# ip nat outside
```

7.3 PAT / Overload Template (Good Extra to Know)

Use this when many inside devices must share one outside IP.

```
access-list 1 permit 192.168.10.0 0.0.0.255
interface fa0/0
 ip nat inside
interface s0/0/0
 ip nat outside
ip nat inside source list 1 interface s0/0/0 overload
```

7.4 NAT Verification

```
show ip nat translations
show ip nat statistics
```

7.5 NAT Rules You Must Not Forget

- Mark the LAN/private side as `ip nat inside`
- Mark the public/transit side as `ip nat outside`
- Static NAT needs a route toward the outside network
- If no traffic has been generated yet, the translation table may be empty

8. NTP: Small Topic but It Can Show Up in Lab 11

```
enable
configure terminal
ntp server 1.0.0.2
end
show clock
show ntp status
```

Meaning: the router becomes an NTP client and syncs time from the server at 1.0.0.2.

9. Mixed Topology Strategy for the Kind of Diagram in Your Photo

When the question gives a bigger hand-drawn topology with many PCs and multiple routers:

1. Number every LAN first. Give each LAN its own network.
2. Put .1 on the router interface for each LAN.
3. Give the PCs .2, .3, .4 and use the router's .1 as the gateway.
4. For every router-router link, use a separate /30 network.
5. Configure all interfaces on all routers first.
6. Only after every interface is up, start RIP or OSPF.
7. If DHCP is asked, do it after routing works.
8. If NAT is asked, do it on the border router after routing works.

Good example network plan for a larger question:

Segment	Network	Example Assignment
LAN A	192.168.10.0/24	Router = 192.168.10.1, PCs = .2, .3
R1-R2	10.10.10.0/30	R1 = 10.10.10.1, R2 = 10.10.10.2
LAN B	192.168.20.0/24	Router = 192.168.20.1, PCs = .2, .3
R2-R3	10.10.10.4/30	R2 = 10.10.10.5, R3 = 10.10.10.6
LAN C	192.168.30.0/24	Router = 192.168.30.1, PCs = .2, .3

10. Fast Troubleshooting by Symptom

Symptom	Most likely cause	Fast fix
Interface says administratively down	Forgot no shutdown	Go into the interface and enter no shutdown
Serial link stays down	No WIC module, wrong cable, no clock on DCE, wrong interface number	Add WIC-2T , use serial DCE, set clock rate 64000 on DCE side, re-check interface names
PC cannot ping its own gateway	Wrong PC IP/mask/gateway or router LAN interface is down	Recheck PC config, router IP, and show ip interface brief
Can ping gateway but not remote LAN	Routing missing	Add static route or fix RIP/OSPF
RIP routes missing from table	Wrong network statement, version mismatch, interface down	Use show ip protocols ; verify both routers advertise correct networks
OSPF neighbors not forming	Wrong wildcard, wrong area, timer mismatch, interface down	Use show ip ospf neighbor ; check area 0 and wildcard masks
PC gets no DHCP address	No DHCP pool, wrong pool network, no helper, no routing, PC not set to DHCP	Check pool, ip helper-address , and client setting
NAT not translating	Inside/outside reversed or missing route	Check ip nat inside , ip nat outside , and routes

11. Command Memory Page

Command	Meaning
enable	Privileged EXEC mode
configure terminal	Global config mode
hostname R1	Name the router
interface fa0/0	Enter interface mode
ip address x.x.x.x y.y.y.y	Set interface IP
no shutdown	Bring interface up
clock rate 64000	Serial DCE clock
router rip	Start RIP
router ospf 1	Start OSPF process 1
ip dhcp pool NAME	Create DHCP pool
ip helper-address x.x.x.x	Relay DHCP to remote server
ip nat inside	Mark private side for NAT
ip nat outside	Mark public side for NAT
ip route net mask next-hop	Static route
show ip route	View routing table
show ip interface brief	View interface summary

12. Lab 10 and Lab 11 What-To-Expect Summary

- **Lab 10:** configure the given network using RIP, then using OSPF, and verify end-to-end pings between PCs.
- **Lab 11:** configure static NAT, do NTP client-server if asked, configure DHCP on a router, and use `ip helper-address` if clients are on another subnet.
- The safest exam order is: **interfaces first, routing second, services third, testing fourth.**

13. Blank Planning Table You Can Fill During the Eval

Device	Interface	IP / Mask	Gateway / Next Hop	Notes
R1				
R1				
R2				
R2				
R3				
PC0				
PC1				
PC2				
PC3				

Last 20-second checklist before you submit:

- Did every used interface get `ip address` and `no shutdown`?
- Did every PC get the correct mask and default gateway?
- On serial links, did only the DCE side get `clock rate 64000`?
- Does `show ip interface brief` show interfaces up/up?
- Does `show ip route` show the remote networks?
- Did you test with `ping` and then save with `copy running-config startup-config`?